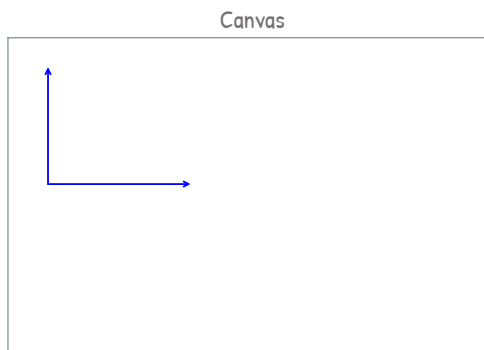
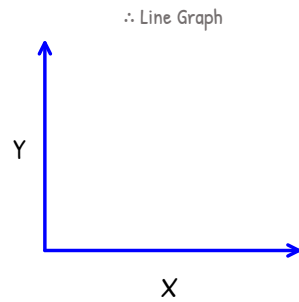
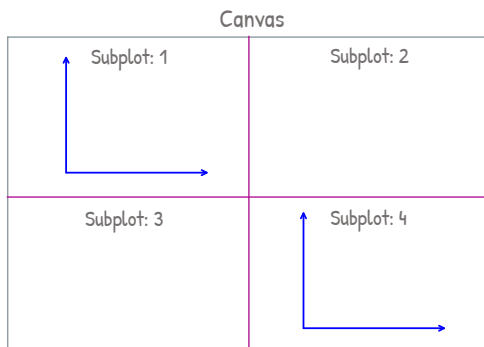




Graphs are drawn on a canvas.

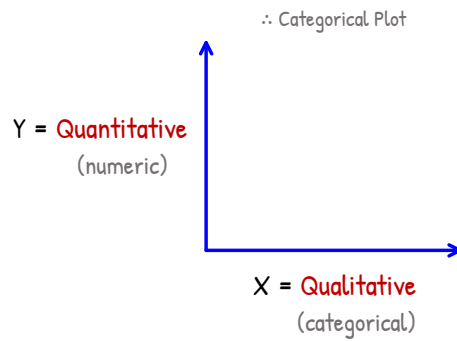
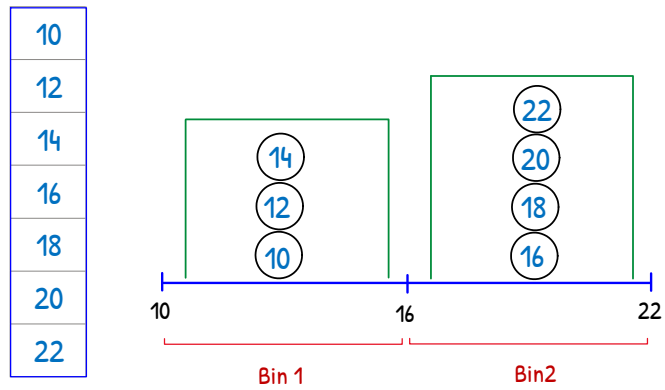
So we can divide canvas into grid to plot multiple graphs over single canvas.



Subplot makes division grid on Canvas

	Age	Salary	Gender
Quantitative (numeric)	22	60k	M
	21	25k	F
	18	30k	M
	27	100k	M
	24	75k	F
			Qualitative (categorical)

About Bins



Let on the basis of A and B
we have to predict C(test).

Correlation

A	B	C
1	1000	100
2	900	200
3	500	300
4	400	450
5	300	570

So if I found any relation between A & C(train) and also B & C(train). then I will use that relation to predict C(test).

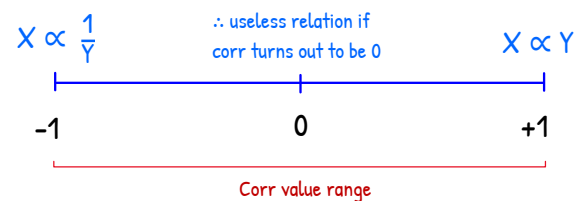
$$A \propto C$$

$$B \propto \frac{1}{C}$$

(A,C) : +ive corr

(B,C) : -ive corr

$$\text{Corr} = \frac{\text{Covariance}}{\text{Standard Deviation}}$$



Chi Square Test

Method 2 : using P-Value

H_0 = No relation

H_1 = Is Relation

```
from scipy.stats import chi2_contingency
```

```
df = sns.load_dataset('titanic')
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
contingency_table = pd.crosstab(df['sex'], df['survived'])
```

survived	0	1
sex		
female	81	233
male	468	109

```
chi2, p_value, dof, expected = chi2_contingency(contingency_table)
```

```
array([[193.47474747, 120.52525253],  
       [355.52525253, 221.47474747]])
```

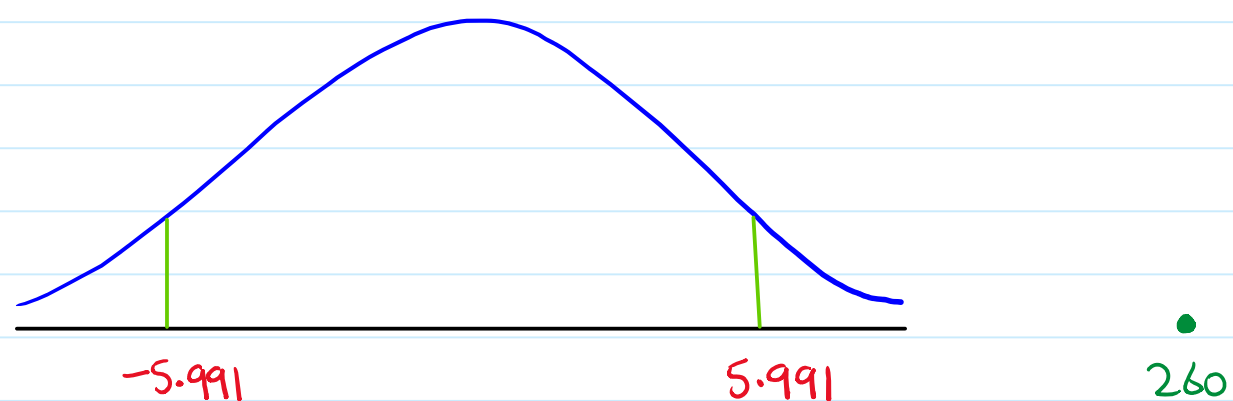
→ Expected Freq.

chi2

```
260.71702016732104
```

p_value

```
1.197357062775565e-58
```



alpha=0.05

If $p_value \leq \alpha$

Reject null hypothesis, there is significant relationship between sex & survived

Else ($p_value > \alpha$)
Accept null hypothesis

ANNOVA TEST

F-test

ANOVA (Analysis of Variance) is used to compare means of more than two independent groups to check if at least one group mean is significantly different.

Annova Test [aka: F-test]

Suppose you want to compare the **average math test scores** of students from **three different teaching methods** (Group A, Group B, Group C).

You want to know:

"Do these teaching methods result in different average scores?"

Method 1: using Critical Value

Critical Method for Annova test is pretty complex
(not that complex but need much calculations)

Also you can find critical value using **Annova Table**.

∴ But here's the catch it has different types of tables so that's why a little tough.
so we are skipping to the code part.

Method 2 : using P-Value

H_0 = No relation

H_1 = Is Relation

```
from scipy.stats import f_oneway
```

```
df = sns.load_dataset('titanic')
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
df = df[['age', 'pclass']].dropna()
```

```
df['pclass'].unique()
```

```
array([3, 1, 2])
```

∴ 3 passenger classes

```
class_1 = df[df['pclass']==1]['age']  
class_2 = df[df['pclass']==2]['age']  
class_3 = df[df['pclass']==3]['age']
```

```
f_stats, p-value = f_oneway(class_1, class_2, class_3)
```

```
f_stats
```

```
57.443484340676214
```

p_value

7.487984171956499e-24

alpha=0.05

If p_value <= alpha

Reject null hypothesis, there is significant
difference in at least one passenger class

Else (p_value > alpha)

Accept null hypothesis



57. —